Coefficient of Friction

Exam code: 9MA0

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Updated on 16 February 2026

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Coefficient of friction

What is friction?

- Friction is a **force** between a surface and an object that acts to resist motion of the object across the surface
- On a horizontal surface with no forces acting on a stationary object other than **gravity** and the surface's **normal reaction force**, there is no force of friction
- If an additional force acts on the object **parallel** to the surface then friction manifests as a force in the **opposite direction**
  - Friction always acts in a direction **parallel** to the **surface**
  - Friction acts in the **opposite direction** to the **resultant force** that would cause the object to move **parallel** to the surface
  - Up to a certain **limiting value** (see next section) the force of friction **prevents motion** entirely
  - Beyond that **limiting value** friction is a force of **constant magnitude** in the **opposite** direction to the motion of the object

What is the coefficient of friction?

- The magnitude of the frictional force satisfies the inequality:
  - $F \leq \mu R$
  - where F is the force of friction

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

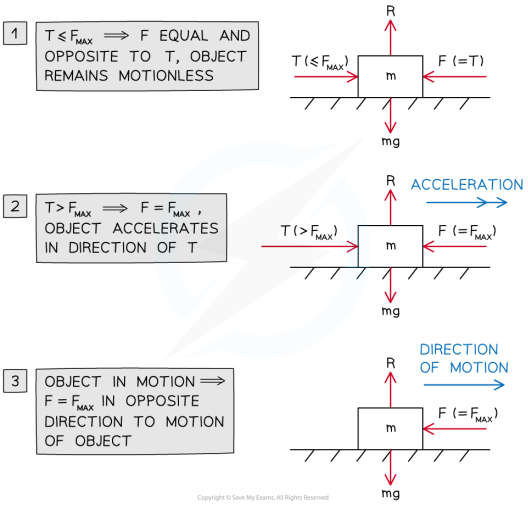
and the surface
    - If there is no friction then $\mu = 0$ and the **surface** is **smooth**
    - Otherwise $\mu > 0$

- The maximum possible frictional force is given by the equation
    - $F_{MAX} = \mu R$
    - where F_{MAX} is the maximum possible force of friction, and R and μ are as above

- F_{MAX} serves as a **limiting value** for mechanics problems involving friction
    - **If the object is stationary** and the resultant of other parallel forces to the surface is **less than or equal to** F_{MAX} then
        - the object **remains stationary**
        - the frictional force is always **equal** and **opposite** to the resultant of the other horizontal forces
        - if the resultant of other horizontal forces is **equal** to F_{MAX} then the object is said to be in **limiting equilibrium**

    - **If the object is stationary** and the resultant of other horizontal forces becomes **greater than** F_{MAX} then
        - the object begins to **accelerate** in the direction of the resultant force
        - the frictional force is equal to F_{MAX} with a direction always opposite to that of the object's motion

    - **If the object is in motion parallel to the surface** then (regardless of any other forces) the frictional force will always be **equal** to F_{MAX} with a direction opposite to that of the object's motion

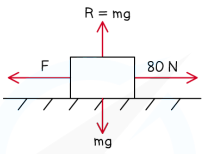


Worked Example

A box with a mass of 12 kg is at rest on a horizontal floor. The coefficient of friction between the box and the floor is 0.7. A

Answer:

START BY SKETCHING A FORCE DIAGRAM!



R = mg = 12 × 9.8 = 117.6 N

SO $F_{MAX} = \mu R = 0.7 \times 117.6 = 82.32$ N

80 N < F_{MAX} , SO THE BOX WILL REMAIN STATIONARY.

THE FORCE OF FRICTION F = 80 N, EQUAL IN MAGNITUDE AND OPPOSITE IN DIRECTION TO THE HORIZONTAL FORCE OF THE BOX.

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Examiner Tips and Tricks

- Always draw a force diagram and label it clearly.
- Look out for the words **smooth** and **rough** in mechanics problems involving an object moving (or potentially moving) along a surface:
  - If the surface is described as **smooth** then you can ignore friction in the problem (ie $\mu = 0$)
  - If the surface is described as **rough** then you need to include the force of friction in solving the problem.
- If a friction question states that an object is **on the point of moving** that means that the object is in **limiting equilibrium.**

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